

NARAYANA

IIT-JEE / NEET / FOUNDATIONS

Class: XI_STU_IC/IR_IIT_B1 & B2

JEE ADVANCED-2022-P2-Model

Date: 17-03-24

Time: 3hrs

PTA-02

Max. Marks: 180

IMPORTANT INSTRUCTIONS

MATHEMATICS					
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 –8)	Questions with Single Digit Integer Answer Type	3	-1	8	24
Sec – II (Q.N : 9 –14)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 15 – 18)	Questions with Single Answer Type	3	-1	4	12
Total				18	60

PHYSICS					
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 –26)	Questions with Single Digit Integer Answer Type	3	-1	8	24
Sec – II (Q.N : 27 –32)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 33 – 36)	Questions with Single Answer Type	3	-1	4	12
Total				18	60

CHEMISTRY					
Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 –44)	Questions with Single Digit Integer Answer Type	3	-1	8	24
Sec – II (Q.N : 45 –50)	Questions with One or more Answer Type	4	-2	6	24
Sec – III (Q.N : 51 – 54)	Questions with Single Answer Type	3	-1	4	12
Total				18	60

Exam Syllabus

TOPIC	
MATHS	Binomial Theorem, Permutations and combinations, Straight line and pair of straight lines, Parabola, Ellipse, Hyperbola
PHYSICS	Mechanical properties of solids, mechanical properties of Fluids, Thermal properties of matter, KTG, Thermodynamics, Oscillations, waves
CHEMISTRY	S-block elements, p-block elements: 13th group elements, 14th group elements Organic Chemistry: Some basic principles and techniques Basic concepts of organic chemistry, Hydrocarbons

SECTION – I

(SINGLE DIGIT INTEGER ANSWER TYPE)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
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1. A parabola with vertex (2, 3) and axis parallel to the y-axis passes through (4, 5). Then its length of latusrectum is
2. M is the foot of the perpendicular from a point P on the parabola $y^2 = 8(x - 3)$ to its directrix and S is the focus of the parabola, if SPM is an equilateral triangle, the length of each side of the triangle is
3. If $(x_1, y_1), (x_2, y_2)$ are the extremities of a focal chord of the parabola $y^2 = 16x$, then $4x_1x_2 + y_1y_2 =$
4. If the coefficients of the 2nd, 3rd and 4th terms in the expansion of $(1 + x)^n$, $n \in \mathbb{N}$, are in AP then n is
5. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is normal to the circle $x^2 + y^2 + 4x + 1 = 0$, then the maximum value of ab is
6. The number of terms with integral coefficients in the expansion of $(7^{1/3} + 5^{1/2}x)^{600}$ is $100 + \lambda$ then λ is
7. If any point on a hyperbola has the coordinates $(5 \tan \phi, 4 \sec \phi)$ then the eccentricity of the hyperbola is 'e' then $[e]$ (where $[.]$ is greatest integer function)
8. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is

SECTION – II

(ONE OR MORE ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has FOUR options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
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Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
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9. The triangle formed by the lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ is
(A) isosceles (B) scalene (C) acute angled (D) obtuse angled

10. If the area of quadrilateral formed by the tangents from the origin to the circle $x^2 + y^2 + 6x - 10y + c = 0$ and the radii corresponding to points of contact is 15, then a value of c is
 (A) 9 (B) 4 (C) 5 (D) 25
11. A quadrilateral is inscribed in a parabola, then
 (A) quadrilateral may be cyclic
 (B) diagonals of the quadrilateral may be equal
 (C) all possible pairs of adjacent sides may be perpendicular
 (D) The quadrilateral may be a square.
12. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$)
 (A) two directrices are parallel to each other at a fix distance of $\frac{2a}{e}$
 (B) distance of focus from its corresponding directrix is $\left| \frac{a}{e} - ae \right|$
 (C) centre is the midpoint of two foci
 (D) distance between vertices is $2ae$
13. x is one among n distinct object and $1 \leq r < n$ an integer. Then which of the following are true?
 (A) The number of permutations of r objects that involve the object x is $r! \binom{n-1}{r-1}$
 (B) The number of permutations of r objects that do not involve the object x is ${}^{n-1}P_r$.
 (C) ${}^nP_r = {}^{n-1}P_r + r \binom{n-1}{r-1}$
 (D) The number of permutations of r objects that involve the object x is $r \binom{n-1}{r-1}$
14. Equation of the tangent to the hyperbola $11x^2 - 4y^2 = 44$ parallel to the line $y = 3x + 4$ is
 (A) $y = 3x + 5$ (B) $y = 3x + 2$ (C) $y = 3x - 2$ (D) $y = 3x - 5$

SECTION – III

(SINGLE ANSWER TYPE)

- This section contains **FOUR (04)** Matching List Sets.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
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15. The number of ways in which 5 boys and 3 girls can sit around a table so that all the girls do not come together is
 (A) 4020 (B) 4120 (C) 4220 (D) 4320

16. The coefficient of x^{50} in $(x + {}^{101}C_0)(x + {}^{101}C_1).....(x + {}^{101}C_{50})$ is
(A) 4^{50} (B) 2^{50} (C) $2^{101} - 1$ (D) 2^{101}
17. The length of latusrectum of the parabola $169((x-1)^2 + (y-3)^2) = (5x - 12y + 17)^2$ is
(A) 1.076 (B) 0.538 (C) 2.154 (D) 4.308
18. Foci of the ellipse $16x^2 + 25y^2 = 400$ are
(A) $(\pm 2, 0)$ (B) $(\pm 3, 0)$ (C) $\left(\pm \frac{24}{5}, 0\right)$ (D) $\left(\pm \frac{48}{5}, 0\right)$



SECTION – I (SINGLE DIGIT INTEGER ANSWER TYPE)

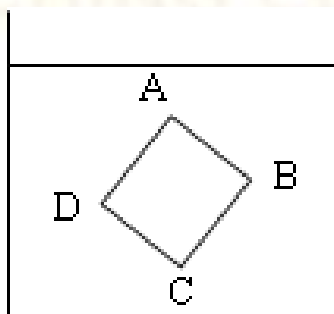
- This section contains **EIGHT (08)** questions.
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19. Pressure at the bottom of tank of water is $3P$, where P is atmospheric pressure. If the water is drawn out till the level of water is lowered by one-fifth, then the pressure at the bottom of the tank is $\frac{13P}{n}$. Find n .
20. A cube of wood supporting a mass of 200 g just floats in water. When the mass is removed, the cube rises by 2 cm. What is $\ell/5$, where ℓ is side length of cube (in cm)? ($\rho_{\text{water}} = 1000 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$)
21. A diatomic gas is compressed adiabatically to $\left(\frac{1}{32}\right)$ of its initial volume. If its initial temperature is T_i (in kelvin) and final temperature is aT_i , the value of a is _____
22. A copper block of mass 4 kg is heated in a furnace to a temperature 425°C and then placed on a large ice block. The mass of ice that will melt in this process will be $0.5x$. Then the closest integer value of x (in kg) is (specific heat copper = $500 \text{ J/kg}^\circ\text{C}^{-1}$ and heat of fusion of ice = 336 kJ/kg)
23. How long after the beginning of motion is the displacement of a harmonically oscillating particle equal to one half its amplitude if the period is 24 s and particle starts from rest (in sec).
24. A body of mass $m = 10 \text{ kg}$ is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station (Breaking stress of wire = $4.8 \times 10^7 \text{ Nm}^{-2}$ and area of cross-section of the wire = 10^{-2} cm^2) is _____
25. A transverse wave pulse is given by $Y = \frac{6}{3 + (3t - 2x)^2}$, where x is in meter and t is in second. If the velocity of the wave pulse is $\frac{n}{2}$ (in m/s), the value of n is _____
26. Two vibrating strings of the same material but lengths L and $2L$ have radii $2r$ and r respectively. They are under the same tension. Both the strings vibrate in their fundamental modes. If frequency of fundamental mode of string of length L is v_1 and the frequency of the fundamental mode of string of lengths $2L$ is v_2 , then find $\frac{v_1}{v_2}$ is _____

SECTION – II
(ONE OR MORE ANSWER TYPE)

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27. The equation of a wave disturbance is given as $y = 0.2 \cos\left(\frac{\pi}{2} + 50\pi t\right) \cos(10\pi x)$, where x and y are in meters and t in seconds. Choose the correct statement(s)
- (A) Antinode occurs at $x = 0.3m$
- (B) The wavelength is $0.2m$
- (C) The speed of the constituent waves is 4 m/s
- (D) Node occurs at $x = 0.15m$
28. A particle is executing SHM with amplitude A . At displacement $x = -A/4$, the magnitude of force acting on the particle is F , potential energy of the particle is U , speed of particle is v and kinetic energy is K . Assume potential energy to be zero at mean position. Then at displacement $x = A/2$, the
- (A) magnitude of force acting on the particle will be $2F$
- (B) potential energy of particle will be $4U$
- (C) speed of particle will be $\sqrt{\frac{4}{5}}v$
- (D) kinetic energy of particle will be $0.8K$
29. The figure shows a container filled with liquid of density ' ρ ' four points A, B, C and D lie on the vertices of a vertical square. Points A and D lie on the vertical plane and points B and C lie on a horizontal plane choose the correct statement (s) about the pressure at the four points



- (A) $P_D = P_B$
- (B) $P_A < P_B = P_D < P_C$
- (C) $P_D = P_B = \frac{P_C - P_A}{2}$
- (D) $P_D = P_B = \frac{P_C + P_A}{2}$

30. If the system is not in free fall, which of the following statements are true about hydrostatic pressure?
- (A) In a liquid, points at different depths can never be at same pressure
 (B) In a liquid, points at different depths may be at same pressure.
 (C) In different liquids, points at same depth can never be at same pressure
 (D) In different liquids, points at different depth can be at same pressure
31. If a sample of metal weighs 210 g in air, 180 g in water and 120 g in liquid
- (A) Relative density of metal is 3
 (B) Relative density of metal is 7
 (C) Relative density of liquid is 3
 (D) Relative density of liquid is $1/3$
32. Three simple harmonic motions in the same direction having the same amplitude and same period are superposed. If each differ in phase from the next by 45° , then
- (A) The resultant amplitude is $(1 + \sqrt{2})a$
 (B) The phase of the resultant motion relative to the first is 90°
 (C) The energy associated with the resulting motion is $(3 + 2\sqrt{2})$ times the energy associated with any single motion.
 (D) The resulting motion is not simple harmonic

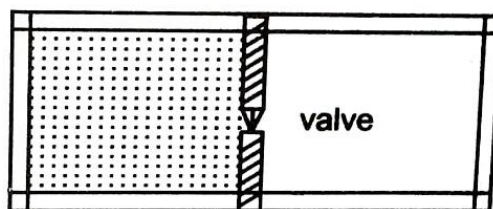
SECTION – III

(SINGLE ANSWER TYPE)

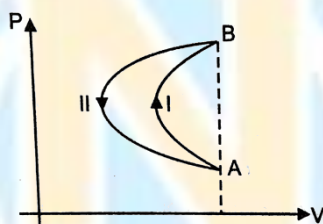
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33. A rod has Poisson's ratio 0.2. If a rod suffers a longitudinal strain of 2×10^{-3} , then the percentage change in volume is
- (A) +0.12 (B) -0.12 (C) 0.28 (D) -0.28

34. A non-conducting container is divided into two chambers that are separated by a valve. The left chamber contains one mole of a monatomic ideal gas. The right chamber is evacuated. At some instant, the valve is opened and the gas rushes freely into the right chamber. Which of the following statements concerning this process is true?



- (A) Work is done by the gas
(B) The temperature of the gas decreases
(C) The walls of the containing vessel must get colder
(D) The change in the internal energy of the gas is zero
35. In a cyclic process, a gas is taken from state A to B via path – I as shown in the indicator diagram and taken back to state A from state B via path – II. In the complete cycle:



- (A) work is done on the gas
(B) heat is given to the gas
(C) not work is done by the gas
(D) nothing can be said about work as data is insufficient
36. Two particles of medium disturbed by the wave propagation are at $x_1 = 0$ and $x_2 = 1 \text{ cm}$. The respective displacements (in cm) of the particles can be given by the equations:

$$y_1 = 2\sin(3\pi t), \quad y_2 = 2\sin(3\pi t - \pi/8)$$

The wave velocity is:

- (A) 16 cm/sec (B) 24 cm/sec (C) 12 cm/sec (D) 8 cm/sec

SECTION – I

(SINGLE DIGIT INTEGER ANSWER TYPE)

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37. How many of the following are heterocyclic aromatic compounds?
Benzene, cyclopentadiene, oxirane, tetrahydrofuran, naphthalene, cyclopentane, tropolone, azulene, biphenyl.
38. Amongst the following the total number of nucleophiles is
 SO_3 , NH_2^- , R^- , H_2O , OR^- , ROH , AlCl_3 , H^- , BF_3 .
39. Number of B-O-B bonds in borax is
40. Amongst the following, the maximum number of compounds which do not behave as Lewis acids are :
 SnCl_2 , H_3BO_3 , AlCl_3 , CF_4 , SiF_4 , SiCl_4 , CCl_4 , BF_3 , SnCl_4 .
41. Total number of elements of group 13 and 14 which form basic oxides are
42. How many silicon atoms are present in the anion of a pyrosilicates?
43. How many of the following s-block elements do not give characteristic colours in the flame test?
Li, Be, Ca, Ba, Sr, Mg, Na, K, Cs.
44. The sulphates of which of the following metals dissolve in water.
 SrSO_4 , K_2SO_4 , BeSO_4 , Li_2SO_4 , MgSO_4 , BaSO_4 , Na_2SO_4 , CaSO_4 , Rb_2SO_4 .

SECTION – II

(ONE OR MORE ANSWER TYPE)

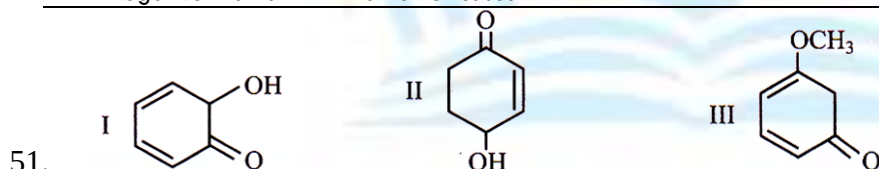
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45. The correct statement is/are:
- | | |
|--|---|
| (A) BeCl_2 is a covalent compound | (B) BeCl_2 is an electron deficient molecule |
| (C) BeCl_2 can form dimer | (D) The hybrid state of Be in BeCl_2 is sp^2 |

46. Which of the following compounds are paramagnetic in nature?
(A) KO_2 (B) K_2O_2 (C) Na_2O_2 (D) RbO_2
47. The compound(s) formed upon combustion of sodium metal in excess air is (are)
(A) Na_2O_2 (B) Na_2O (C) NaO_2 (D) NaOH
48. In acidic medium, the reaction of H_2O_2 with potassium permanganate produces a compound in which the oxidation state of Mn is not?
(A) 0 (B) +2 (C) +3 (D) +4
49. The compound(s) which have -O-O-bond(s) is/are
(A) BaO_2 (B) Na_2O_2 (C) CrO_5 (D) Fe_2O_3
50. The Lewis acid strength of following molecule is in the order
(A) $\text{SiF}_4 < \text{SiCl}_4 < \text{SiBr}_4$ (B) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3$
(C) $\text{SiF}_4 > \text{SiCl}_4 > \text{SiBr}_4$ (D) $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3$

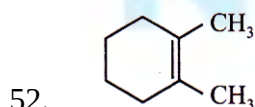
SECTION – III (SINGLE ANSWER TYPE)

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Which among these can exhibit tautomerism?

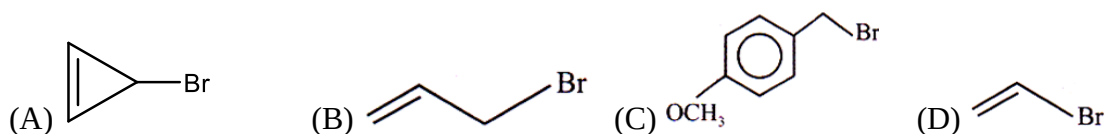
- (A) I only (B) II only (C) III only (D) All of these

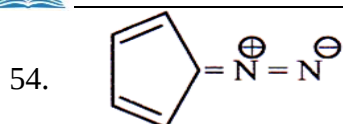


How many geometrical isomers of this compound are possible which can be isolated at room temperature?

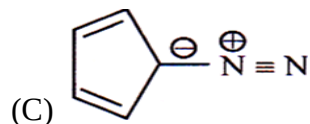
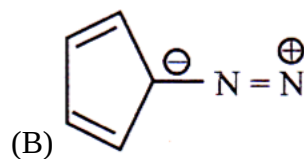
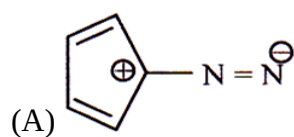
- (A) 0 (B) 2 (C) 3 (D) 4

53. Ease of ionization to produce carbocation and bromide ion under the treatment of Ag^+ will be maximum in which of the following compounds?





The most stable canonical structure of this molecule is



(D) All are equally stable

Subject	Physics	Chemistry	Mathematics
Setter Branch	DEL-PB-CC-162		
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NARAYANA
COACHING CENTER